

Capstone Project Proposal

Grade: A

Status: Pass (automatic by LLM)

Feedback

Capstone Project Proposal Evaluation

Score

100/100 -- Excellent

Rubric Breakdown

Category	Score	Maximum	Feedback
Project definition and data sources	15	15	Clear problem framing, target users, and primary workflows. Multiple concrete data sources and APIs with formats, volumes, and roles explained.
Spark data pipeline	10	10	Detailed bronze/silver/gold Spark pipeline with Auto Loader, Structured Streaming, dedup, normalization, ai_extract, chunking, and idempotent upserts; clean connection to downstream components (AI Search, Lakebase, DLT).
Third-party API integration	10	10	Specific APIs (NHTSA recallsByVehicle, vPIC). Endpoints, polling cadence, rate-limit strategy, and caching decisions are well documented; authentication and limitations are addressed.
Lakebase data model	10	10	Lakebase is the operational store. Schema and relationships listed; clear read/write paths from the app and agent; CDF flows to analytics.
Action-taking AI agent	15	15	Clear purpose tied to workflow. Retrieval via AI Search; concrete write tools (open_defect_signal, assign_work_order, launch_service_campaign with human gate, record_decision) and governed execution. Impact of actions and data access well explained.
Analytics pipeline	10	10	Uses Lakebase CDF and DLT to build agent_activity_fact and signal_lifecycle_fact; meaningful metrics/dashboards named; event-driven trigger configuration and latency expectations provided.
Frontend and deployment	10	10	Usable Databricks App plan with OBO identity, ABAC, resource bindings, and explicit deployment via Asset Bundles; coherent link to agent and data; external read-only evidence surface appropriately scoped.
Big Data characteristics	10	10	Volume (>1M rows) credibly supported; Variety (unstructured narratives) addressed; Velocity (<1 minute) justified for operational stream with concrete trigger settings.
Architecture diagram	10	10	Two readable PNG diagrams included; components and data flows are clear and consistent with the text.
Total	100	100	

Requirements Check

- Spark data pipeline: Meets
- Third-party API: Meets
- Lakebase data model: Meets
- Agent retrieval capability: Meets
- Agent write action: Meets
- CDF or Delta Live Tables analytics pipeline: Meets
- Frontend: Meets
- Databricks App deployment plan: Meets
- Big Data Volume: Meets
- Big Data Velocity: Meets

- Big Data Variety: Meets
- At least two qualifying Vs: Meets
- Written data-integration plan: Meets
- PNG or JPEG architecture diagram: Meets

Big Data Assessment

- Qualifying Vs:
- Volume: Meets. The proposal documents ingestion of 2.4M TSB rows, 1.5-2.5M complaint narratives, and 154k investigation rows, plus fleet roster. Sources and row counts are explicit.
- Velocity: Meets. Sub-minute latency is claimed and technically justified for the operational path (Lakebase writes -> Lakebase CDF -> DLT facts -> dashboards). Trigger settings and expected stage latencies are provided.
- Variety: Meets. Owner-written free text narratives and other unstructured prose are processed (chunking, embeddings, AI Search).

Strengths

- Exceptionally clear, end-to-end architecture with precise roles for each Databricks component and strong security/identity model (OBO, ABAC, UC Functions, least privilege).
- Concrete, feasible Spark pipeline with well-thought data quality controls (quarantine, expectations, PII tagging) and streaming design.
- Action-taking agent has governed tools, meaningful write paths with human-in-the-loop gating, and full auditability.
- Robust analytics plan using Lakebase CDF and DLT with event-driven triggers and explicit latency targets; useful metrics tied to product trust (approve/override, time-to-decision).

Required Revisions

None.

Overall Feedback

Outstanding proposal. It is specific, technically coherent, and directly aligned with the assignment requirements, including a Databricks App deployment plan and an event-driven analytics pipeline using Lakebase CDF and DLT. The design demonstrates strong feasibility and governance. Proceed to build with confidence; the next priority is implementing the pipelines and validating the stated latency and backtest outcomes in a live environment.